Melanoma and Neural System Tumor Syndrome (Melanoma–Astrocytoma Syndrome): A Comprehensive Disease Report

MONDO: MONDO:0007967 · **OMIM:** 155755 · **Causal gene:** *CDKN2A* (9p21.3) · **Inheritance:** Autosomal dominant · **Category:** Mendelian cancer-predisposition syndrome

Summary

Melanoma and Neural System Tumor Syndrome—more commonly called **melanoma–astrocytoma syndrome (MAS)**—is a rare, autosomal-dominant hereditary cancer-predisposition syndrome caused by **germline loss-of-function variants in the *CDKN2A* tumor-suppressor gene at chromosome 9p21.3**. The defining clinical feature is the co-occurrence, within families and often within individuals, of **cutaneous melanoma** and **tumors of the neural/nervous system** (most characteristically astrocytoma and other gliomas, but also glioblastoma, schwannoma, and meningioma). The *CDKN2A* locus is unusual in that it encodes **two distinct tumor-suppressor proteins from overlapping reading frames**: **p16INK4A**, which restrains the CDK4/6–Rb cell-cycle checkpoint, and **p14ARF** (p19ARF in mouse), which stabilizes p53 by inhibiting MDM2. This dual-protein architecture explains the dual-tumor phenotype: loss of p16INK4A predisposes principally to melanoma, whereas loss of **p14ARF (encoded by the ARF-specific exon 1β)** is the critical lesion underlying the neural-system tumor component.

The mechanistic core of the disease is a **two-hit tumor-suppressor model**: affected individuals inherit one inactivated *CDKN2A* allele in the germline, and tumors arise when the remaining wild-type allele is somatically lost (commonly by homozygous 9p21 deletion). In melanocytes, loss of p16INK4A allows cells carrying oncogenic drivers (e.g., BRAF-V600E) to **escape oncogene-induced senescence**, enabling progression from nevus to melanoma. In glial and neural-crest-derived cells, loss of p14ARF **disables the ARF–MDM2–p53 axis**,

removing a key apoptotic/senescence barrier and cooperating with additional somatic drivers (e.g., BRAF alterations) to produce astrocytomas. Somatic **homozygous *CDKN2A/B* deletion** is now a formally recognized molecular marker of grade 4 in IDH-mutant astrocytoma under the 2021 WHO CNS classification.

Penetrance is **incomplete and age-dependent**, with lifetime melanoma risk estimates of 52–84% for high-penetrance carriers, and is strongly modulated by **environmental factors (UV radiation for melanoma; tobacco smoking for pancreatic cancer)** and **modifier genes (notably MC1R)**. *CDKN2A* carriers also carry a substantially elevated risk of **pancreatic ductal adenocarcinoma (PDAC)**. Clinical management is not syndrome-specific but combines **germline genetic testing and counseling, UV/smoking risk reduction, dermatologic surveillance, and pancreatic imaging surveillance (MRI/MRCP ± EUS)**, with tumors treated per standard melanoma, glioma, and pancreatic-cancer protocols. The *CDKN2A*–CDK4/6 axis is a rational precision-oncology target (CDK4/6 inhibitors).

Disease Information (Section 1)

Overview. Melanoma–astrocytoma syndrome is an inherited syndrome in which individuals have an increased risk of developing both **cutaneous malignant melanoma** and **nervous-system tumors**, classically astrocytomas. It represents a phenotypic subset of the broader *CDKN2A*-related familial melanoma / familial atypical multiple mole melanoma (FAMMM) spectrum, distinguished by the presence of neural-system tumors. Because melanocytes and glial/Schwann cells share a **common neuroectodermal/neural-crest embryologic origin**, a shared genetic lesion predisposing to both is biologically coherent (PMID: 36980355).

Key identifiers:

Resource	Identifier
MONDO	MONDO:0007967
OMIM	155755
Causal gene	*CDKN2A* (HGNC:1787), 9p21.3
Synonyms	Melanoma–astrocytoma syndrome; melanoma and neural system tumor syndrome; CMM+NST (cutaneous malignant melanoma + neural system tumor)

Synonyms / alternative names: melanoma–astrocytoma syndrome; melanoma–neural system tumour syndrome; CMM+NST syndrome.

Information source type. The evidence base derives overwhelmingly from **aggregated disease-level resources**—individual case reports, melanoma-prone family kindreds, familial-melanoma registries (e.g., the Dutch national familial melanoma database), and biobank-scale cohorts (UK Biobank, Geisinger MyCode)—rather than from single-patient EHR extraction.

Etiology (Section 2)

Primary cause (genetic). The disease is caused by **germline inactivating variants in *CDKN2A***. Multiple kindreds demonstrate germline 9p21 deletions or point mutations spanning *CDKN2A*. Bahuau et al. 1998 found a lesion disrupting p16 and p19ARF but sparing p15 that segregated with melanoma–astrocytoma syndrome (MIM 155755) (PMID: 9622062). Critically, Randerson-Moor et al. 2001 reported a family with a **germline deletion of the p14ARF-specific exon 1β only** (not affecting p16 or CDKN2B coding regions), implicating **p14ARF loss as the critical driver of the neural-system tumor component** (PMID: 11136714). Exon 1β / ARF-specific lesions predisposing to CNS tumors are corroborated by Wong 2022 (PMID: 35904890) and Hewitt 2002 (PMID: 12019208).

Genetic risk factors. - *CDKN2A* germline loss-of-function variants (causal). - **CDK4** germline mutations cause a smaller fraction of hereditary melanoma (functionally equivalent downstream lesion) (PMID: 10630172). - **MC1R** (melanocortin-1 receptor) variants act as low-penetrance **modifiers** that increase melanoma risk and cosegregate with deleterious p16 alleles (PMID: 16462187).

Environmental risk factors. - **Ultraviolet (UV) radiation** is the dominant environmental driver of the melanoma component. "The penetrance of these melanoma-predisposing genes is largely influenced by ultraviolet exposure across geographic latitude" (PMID: 16462187). - **Tobacco smoking** dramatically amplifies pancreatic-cancer risk in carriers (HR 25.8 for ever-smokers vs non-carriers; $p = 2.1 \times 10^{-13}$) (PMID: 21150883). - **Age, sex, and family history** act as standard modifiers; penetrance rises steeply with age.

Protective factors. - **Environmental:** UV avoidance / sun protection and smoking cessation reduce melanoma and pancreatic risk respectively. Genetic counseling measurably reduces objectively measured UV exposure in carriers (PMID: 31371819). - **Genetic protective factors:**

No well-established protective alleles are documented specific to this syndrome; absence of high-risk MC1R variants and a darker constitutive pigmentation background are relatively protective against the UV-driven melanoma component.

Gene–environment interactions. Two strong G×E interactions are documented: (1) **UV × CDKN2A** for melanoma—UV induces melanoma-like cell-cycle/DNA-damage transcriptomic changes even in phenotypically normal *CDKN2A*-heterozygous fibroblasts (PMID: 23371019); and (2) **smoking × CDKN2A** for pancreatic cancer (HR 25.8) (PMID: 21150883).

Phenotypes (Section 3)

The syndrome is defined by a **dual tumor spectrum**. In an unbiased UK Biobank cohort of MAS cases, the tumor spectrum was: brain neoplasms 33% (glioblastoma, gliosarcoma, astrocytoma), schwannomas 25%, cutaneous melanoma 17%, and head/neck squamous cell carcinoma 17% (PMID: 37585199).

Phenotype	Type	Frequency	Onset / course	Suggested HPO term
Cutaneous melanoma	Neoplasm (physical manifestation)	~17% (biobank); lifetime risk 52–84% in high-penetrance carriers	Younger age at onset than sporadic; often multiple primaries	HP:0012056 (Cutaneous melanoma)
Multiple primary melanomas	Neoplasm	Up to ~10% of melanoma series	Earlier onset; carriers	HP:0012056
Dysplastic/atypical nevi	Physical manifestation (precursor)	Frequent in kindreds	Childhood–adult	HP:0003764 (Nevus)
Astrocytoma / diffuse glioma	Neoplasm (nervous system)	Major neural component	Variable; pediatric or adult	HP:0009592 (Astrocytoma)
Glioblastoma / gliosarcoma	Neoplasm (nervous system)	Within the 33% brain-tumor group	Adult; aggressive	HP:0100843 (Glioblastoma multiforme)
Schwannoma / peripheral nerve sheath tumor	Neoplasm (nervous system)	~25%	Variable	HP:0100008 (Schwannoma)
Meningioma	Neoplasm (nervous system)	Reported	Adult	HP:0002858 (Meningioma)
Pancreatic ductal adenocarcinoma	Neoplasm (digestive)	~8.9% cumulative in surveilled carriers; up to 58% penetrance by age 80	Adult/late	HP:0002894 (Pancreatic adenocarcinoma)
NF1-like features			—	—

Phenotype	Type	Frequency	Onset / course	Suggested HPO term
	Physical manifestation	Reported subset		

Severity and progression. Phenotype is **variable** with incomplete penetrance and variable expressivity. Melanomas in carriers tend to be thinner (lower Breslow thickness), superficial-spreading, and in-situ predominant, consistent with younger onset and surveillance detection (PMID: 30218143). Neural tumors range from low-grade astrocytoma to lethal glioblastoma.

Quality-of-life impact. No syndrome-specific QoL instruments (EQ-5D/SF-36) data were identified. QoL impact derives from the component tumors—neurological deficit, seizures, and cognitive effects from brain tumors; surgical morbidity; and the psychological burden of lifelong cancer surveillance.

Genetic / Molecular Information (Section 4)

Causal gene. *CDKN2A* (cyclin-dependent kinase inhibitor 2A; HGNC:1787; OMIM 600160), located at **9p21.3**. The *CDKN2A* locus uses alternative first exons and overlapping reading frames to encode two unrelated proteins: - **p16INK4A** (exon 1α) — CDK4/6 inhibitor acting in the Rb pathway. - **p14ARF** (exon 1β; p19ARF in mouse) — MDM2 inhibitor acting in the p53 pathway.

The neighboring **CDKN2B** gene (encoding p15INK4B) is frequently co-deleted in large 9p21 deletions.

Pathogenic variant classes. - **Large germline 9p21 deletions** spanning p16/p14ARF (± CDKN2B) (PMID: 9622062). - **ARF-specific (exon 1β) deletions / point mutations**—e.g., germline exon 1β deletion (PMID: 11136714); exon 1β deletion with melanoma + neural tumors (PMID: 35904890); an ARF splice mutation causing ARF haploinsufficiency (PMID: 12019208); p14ARF exon 1β missense G16D in a CMM+NST family (PMID: 15937071). - **Founder frameshift variants — p16-Leiden (c.225_243del, p.Ala76fs)**, the major Dutch founder allele (PMID: 32482799, PMID: 22636603); and c.19_23dup (p.Ser8fs). - **Missense variants** — e.g., c.442G>A (p.Ala148Thr), c.249C>A (p.His83Gln) (PMID: 32221274).

Variants are classified **pathogenic/likely pathogenic** per ACMG/AMP for the recurrent founder and truncating alleles; some missense changes (e.g., p.Ala148Thr) are of debated/variable clinical significance.

Somatic vs germline. The syndrome is **germline-driven** (one inherited hit), with tumors completing inactivation via **somatic loss of the second allele**. In MAS astrocytomas, tumors showed **homozygous deletion of *CDKN2A/B* due to somatic loss of the other copy of chromosome 9p**, plus additional somatic drivers (BRAF) (PMID: 28699883).

Functional consequence. Loss of function of both tumor-suppressor proteins (p16INK4A and p14ARF), producing failure of cell-cycle arrest and of p53-mediated senescence/apoptosis.

Modifier genes. MC1R (melanoma risk modifier) (PMID: 16462187); **MITF** (e.g., p.Glu318Lys, associated with nodular melanoma) (PMID: 32221274).

Epigenetic information. While MAS is primarily driven by genetic *CDKN2A* loss, the associated IDH-mutant gliomas display extensive **DNA/histone methylation reprogramming (G-CIMP)**; DNA methylation profiling is used for CNS tumor classification, integrated with homozygous *CDKN2A/B* deletion status (PMID: 42630077).

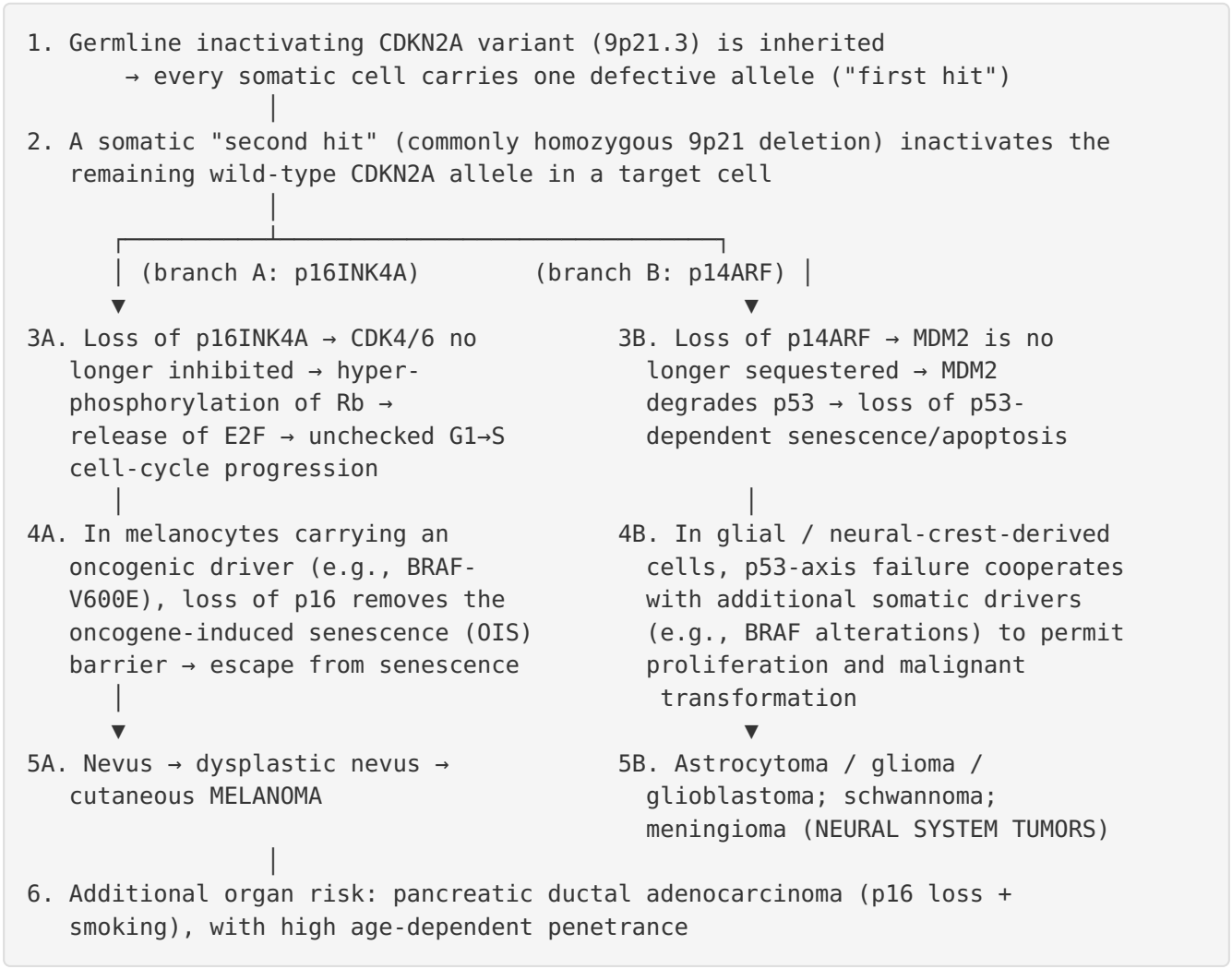
Chromosomal abnormalities. Large-scale **9p21 germline deletions** are the hallmark structural lesion; tumors frequently show **9p21 homozygous deletion** and, in IDH-wildtype glioblastoma, chromosome +7/−10 (PMID: 42158566).

Environmental Information (Section 5)

- **Environmental factors: UV radiation** (solar and artificial) is the principal environmental carcinogen driving the melanoma component; its effect is strongly latitude-dependent and interacts with *CDKN2A* and MC1R genotype (PMID: 16462187). UV induces melanoma-like transcriptomic dysregulation in *CDKN2A*-heterozygous cells (PMID: 23371019).
- **Lifestyle factors: Tobacco smoking** is a powerful modifier of pancreatic-cancer risk in carriers (HR 25.8) (PMID: 21150883). Sun-seeking behavior and tanning-bed use increase melanoma risk.
- **Infectious agents:** None. This is a Mendelian cancer-predisposition syndrome with no infectious etiology.

Mechanism / Pathophysiology (Section 6)

Ordered causal chain



(Branches A and B are both engaged when large germline deletions disrupt the whole locus; ARF-specific exon-1β lesions preferentially engage branch B and the neural component. Steps 4A/4B are mechanistically demonstrated in vitro and in mouse; the exact cell-of-origin sequence for the human neural tumors is partly inferred.)

Detail

Molecular pathways. Two canonical tumor-suppressor pathways converge at the *CDKN2A* locus: - **p16INK4A → CDK4/6 → Rb → E2F (cell-cycle control).** "The p16INK4a tumor suppressor protein functions as an inhibitor of CDK4 and CDK6, the D-type cyclin-dependent

kinases that initiate the phosphorylation of the retinoblastoma tumor suppressor protein, RB" (PMID: 15460900). Loss of p16 releases the Rb checkpoint, permitting uncontrolled G1→S transition. - **p14ARF → MDM2 → p53 (apoptosis/senescence control).** "The role of ARF as a tumor suppressor is closely linked to the p53-MDM2 axis, which is a key process that restrains tumor formation" (PMID: 29665672). ARF sequesters MDM2, stabilizing p53; ARF loss leads to MDM2-mediated p53 degradation. - **MAPK (BRAF) cooperation.** Oncogenic BRAF/RAS signaling provides the proliferative drive that *CDKN2A* loss un-brakes; in glioma, p53-pathway disablement by p14ARF deletion combines with MDM2/MDM4 amplification (PMID: 21200141).

Cellular processes. The central cellular mechanism is **failure of oncogene-induced senescence (OIS)**. "Malignant melanomas often arise from nevi, which result from initial oncogene-induced hyperproliferation of melanocytes that are maintained in a CDKN2A/p16-mediated senescent state" (PMID: 27872202). Removal of the p16 barrier (or CDK6 overexpression) bypasses senescence; removal of the ARF–p53 barrier removes apoptotic backup. Both result in **cell-cycle dysregulation** and escape from two independent proliferation brakes.

Protein dysfunction. Both encoded proteins are rendered non-functional (**loss of function**): p16INK4A can no longer bind/inhibit CDK4/6; p14ARF can no longer bind/sequester MDM2.

Immune / metabolic / tissue-damage mechanisms. UV-induced DNA damage in melanocytes is the proximal mutagenic insult for the melanoma branch; in the pancreatic branch, tobacco carcinogens plus p16 loss drive adenocarcinoma. These are **genotoxic/oxidative-stress mechanisms** rather than autoimmune or metabolic.

Suggested ontology terms. - GO biological process: GO:0007050 (cell cycle arrest); GO:0090398 (cellular senescence); GO:0006915 (apoptotic process); GO:0072332 (intrinsic apoptotic signaling by p53 class mediator); GO:0000079 (regulation of CDK activity). - CL cell types: CL:0000148 (melanocyte); CL:0000127 (astrocyte); CL:0002573 (Schwann cell); neural-crest-derived lineages.

Anatomical Structures Affected (Section 7)

Organ level. - **Primary:** skin (UBERON:0002097), brain (UBERON:0000955), peripheral nerves (UBERON:0001021). - **Secondary:** pancreas (UBERON:0001264); head/neck mucosa (squamous cell carcinoma reported). - **Body systems:** integumentary, central nervous system, peripheral nervous system, digestive (pancreas).

Tissue and cell level. - **Tissues:** epidermal/melanocytic (skin), glial/neural tissue (brain), peripheral nerve sheath (connective/neural). - **Cell populations (CL terms):** melanocytes (CL:0000148), astrocytes (CL:0000127), Schwann cells (CL:0002573), pancreatic ductal epithelial cells (CL:0002079). All melanocytes, astrocytes, and Schwann cells share a **neural-crest / neuroectodermal origin**, which underlies the syndrome's combined skin + nervous-system phenotype (PMID: 36980355).

Subcellular level (GO Cellular Component). Nucleus (GO:0005634) — site of Rb/E2F and p53/MDM2 regulation; nucleolus (GO:0005730) — where ARF resides and senses nucleolar/oncogenic stress; cytoplasm for MDM2 shuttling.

Localization / lateralization. Melanomas arise at sun-exposed cutaneous sites; brain tumors are commonly supratentorial but can occur in posterior fossa. Lateralization is tumor-specific and not characteristically unilateral or bilateral at the syndrome level.

Temporal Development (Section 8)

- **Onset.** Variable. Melanoma in *CDKN2A* carriers typically presents at a **younger age than sporadic melanoma**, often with multiple primaries (PMID: 32221274, PMID: 30218143). Neural tumors can present in childhood or adulthood. Pancreatic cancer is adult/late-onset.
- **Progression.** The melanoma branch follows the **nevus → dysplastic nevus → melanoma** sequence. Astrocytomas can progress to higher grade; homozygous *CDKN2A/B* deletion upgrades IDH-mutant astrocytoma to WHO grade 4 (PMID: 42630077). Progression rate is variable and tumor-dependent.
- **Disease course.** Lifelong predisposition (chronic, germline) with episodic tumor events; penetrance accumulates with age.
- **Critical periods.** Cumulative UV exposure throughout life is the key modifiable window for the melanoma branch; genetic counseling can intervene behaviorally (PMID: 31371819).

Inheritance and Population (Section 9)

Epidemiology. MAS is **rare**. Sargen et al. 2023 estimated prevalence ranging from **1 in 170,503 (Geisinger MyCode)** to **1 in 39,149 (UK Biobank; 12 cases / 469,789)** (PMID: 37585199).

Inheritance pattern. Autosomal dominant with incomplete, age-dependent penetrance and variable expressivity. The Dutch national familial melanoma database identified 172 families carrying 15 different pathogenic *CDKN2A* variants in 649 individuals (PMID: 32482799).

Penetrance. Incomplete and age-dependent. For *CDKN2A* carriers, penetrance by age 80 was estimated at **58% for pancreatic cancer (95% CI 8–86%)** and **39% for melanoma (95% CI 0–80%)** in one unselected pancreatic-cancer-ascertained cohort (PMID: 21150883); high-penetrance carriers in familial-melanoma series face lifetime melanoma risk of **52–84%** (PMID: 38287743).

Expressivity. Variable—some families express predominantly melanoma, others melanoma + neural tumors + pancreatic cancer.

Founder effects. p16-Leiden (c.225_243del, p.Ala76fs) is the predominant Dutch founder allele (PMID: 32482799, PMID: 22636603).

Genetic anticipation / germline mosaicism / consanguinity: no specific evidence; not a repeat-expansion disorder.

Population demographics. Documented in multiple populations (Dutch, Italian, UK). Melanoma penetrance is higher in high-UV, low-latitude geographies (PMID: 16462187). No strong sex ratio established at the syndrome level.

Diagnostics (Section 10)

Genetic testing (the cornerstone). Diagnosis rests on identifying a **germline *CDKN2A* pathogenic variant**, by single-gene testing or multigene hereditary-cancer panels. A *CDKN2A* mutation is identified in **~10–40% of families meeting testing criteria**, and multigene panels increase yield (PMID: 38287743). Testing thresholds are incidence-dependent: ≥2 affected first-degree relatives in low-incidence countries, 3–4 affected members in high-incidence countries (USA/Australia) (PMID: 38287743). A diagnosis of MAS specifically requires a germline *CDKN2A* pathogenic variant **plus ≥1 neural-system tumor** (PMID: 37585199). WES/WGS and gene panels are all applicable; chromosomal microarray / FISH can detect large 9p21 deletions.

Tumor (somatic) diagnostics. - **Homozygous *CDKN2A/B* deletion** on tumor sequencing, FISH, or chromosomal microarray—a molecular criterion for grade-4 IDH-mutant astrocytoma (PMID: 42630077, PMID: 28699883). - **DNA methylation profiling** for CNS-tumor classification

(PMID: 42630077). - **Histopathology / immunohistochemistry:** loss of p16 staining supports tumor diagnosis; **S100B** and dermoscopic/histologic criteria (ABCDE) aid melanoma diagnosis (PMID: 40581420).

Clinical criteria. Dermatologic examination for atypical/dysplastic nevi; neuroimaging (MRI) for neural tumors. **Differential diagnosis** includes other familial-melanoma genes (CDK4, MITF, BAP1, POT1), neurofibromatosis type 1 (MAS can have NF1-like features), and Li-Fraumeni syndrome.

Screening. Cascade genetic testing of relatives; dermatologic surveillance; pancreatic imaging surveillance in carriers (below).

Outcome / Prognosis (Section 11)

Melanoma. Carrier melanomas tend to be **thinner and earlier-stage** (more in-situ and superficial-spreading, lower Breslow thickness), consistent with surveillance detection and favorable early-stage prognosis (PMID: 30218143). Modern adjuvant therapy improves outcomes: 5-year melanoma-specific survival of **74% with adjuvant therapy vs 60% with observation (adjusted HR 0.58)** in high-risk resected melanoma (PMID: 42849920).

Neural tumors. Prognosis depends on tumor type/grade—astrocytomas carry variable prognosis; glioblastoma is highly lethal. Homozygous *CDKN2A/B* deletion marks aggressive (grade-4) IDH-mutant astrocytoma (PMID: 42630077). High-grade astrocytoma with piloid features (CDKN2A/B loss in 78%) had median overall survival of ~88 months (PMID: 42747321).

Pancreatic cancer. Historically lethal (sporadic 5-year survival ~8%), but **surveillance improves outcomes**: in 347 *CDKN2A* carriers followed 20 years, PDAC developed in 31 (8.9%), with **83.3% resectable at imaging**, and **5-year survival 32.4% overall / 44.1% after resection** (PMID: 35658523). Surveillance reduced PDAC mortality by 12.1% and increased life expectancy by 2.10 years (PMID: 36785917).

Prognostic factors / biomarkers. Tumor stage/grade, molecular markers (homozygous *CDKN2A/B* deletion, IDH status, ATRX), resectability of pancreatic lesions, Breslow thickness for melanoma.

Treatment (Section 12)

There is **no syndrome-specific therapy**; component tumors are treated per standard oncology protocols, but the *CDKN2A*–CDK4/6 axis is an actionable precision target.

Melanoma (NCIT: C15262 immunotherapy; C15632 targeted therapy). - **Immune checkpoint inhibitors (anti-PD-1)** and **BRAF/MEK inhibitors** substantially improve survival in advanced melanoma (PMID: 42891779, PMID: 42907470). - **Adjuvant anti-PD-1 / BRAF-MEK** improves 5-year melanoma-specific survival (74% vs 60%, HR 0.58) (PMID: 42849920). - Molecular tumor-board profiling identifies **CDK4/6 pathway amplifications** as actionable targets (PMID: 42907470).

CDK4/6 inhibition (precision target; NCIT: C178229 class). Because *CDKN2A/p16* loss causes **universal activation of CDK4/6 and Rb pathways**, CDK4/6-specific inhibitors (e.g., **palbociclib**) are a rational targeted strategy in CDKN2A-null tumors (PMID: 26183925).

Neural tumors. Astrocytomas: maximal safe surgical resection, radiotherapy, and alkylating chemotherapy (temozolomide). For WHO grade-2 IDH-mutant gliomas, **vorasidenib** (mutant-IDH inhibitor) can delay progression in selected patients (PMID: 42430039).

Pancreatic cancer. Surgical resection when detected early via surveillance; standard systemic chemotherapy; PARP inhibitors for BRCA-associated cases (not *CDKN2A*-specific).

Pharmacogenomics / personalized medicine. BRAF-V600 status guides BRAF/MEK inhibitor use; genotype-guided CDK4/6 inhibition is under investigation.

Prevention (Section 13)

- **Primary prevention:** UV risk reduction (sun protection, avoidance of tanning beds) and smoking cessation. Monthly skin self-examination and sun-exposure moderation are long-standing recommendations (PMID: 10630172). Genetic counseling with test disclosure produces measurable reductions in objectively measured UV exposure and increases in skin pigmentation one year later (PMID: 31371819); priority-of-risk perception predicts improved sun protection (PMID: 32415830).
- **Secondary prevention:** Dermatologic surveillance (clinical skin exams 1–2×/year) detects early curable melanoma (e.g., a curable melanoma detected in a 17-year-old during surveillance) (PMID: 22636603). **Pancreatic surveillance** with annual MRI/MRCP ±

endoscopic ultrasound detects resectable PDAC and is cost-effective (€14,000/QALY) (PMID: 35658523, PMID: 36785917).

- **Tertiary prevention:** Standard tumor-directed therapy and ongoing surveillance for additional primaries.
- **Genetic counseling / cascade testing:** central to risk assessment and family planning; all kindred members (regardless of mutation status) should adhere to sun protection and skin surveillance (PMID: 30775140).
- **Immunization / public health / infectious control:** not applicable.

Other Species / Natural Disease (Section 14)

- **Orthologous gene:** mouse *Cdkn2a* (Ink4a/Arf locus), NCBI Gene ID 12578; the mouse ARF protein is **p19ARF** (vs human p14ARF).
- **Natural disease:** No well-characterized naturally occurring melanoma–astrocytoma syndrome in companion animals or wildlife was identified in this investigation; the disease is studied primarily through engineered mouse models (below).
- **Comparative biology:** The Ink4a/Arf tumor-suppressor architecture and both the Rb and p53 pathways are **evolutionarily conserved** between mouse and human, making the mouse a faithful system for the core mechanism (PMID: 12406321).
- **Transmission / zoonotic potential:** not applicable (germline genetic disease).

Model Organisms (Section 15)

Mouse models recapitulate *CDKN2A*-driven melanoma predisposition via Ink4a/Arf loss cooperating with Ras/MAPK activation and UV: - "Both the Ink4a/pRb and Arf/p53 pathways are involved in melanoma development in mice" (PMID: 12406321); Cdkn2a (Ink4a/Arf) loss cooperates with Ras/MAPK activation to induce melanoma. - **HGF:p16/Cdkn2a UV-induced melanoma GEMM** used to study metastasis suppressors (Nme1/Nme2) (PMID: 33024267). - **ESC-GEMM platforms** combine Cdkn2a loss with Braf/Pten alleles for rapid melanoma modeling (PMID: 31744817). - **Knockout mice** for CDK inhibitors generated by engineered-nuclease genome editing (PMID: 30671114).

Model types: knockout, conditional, transgenic, and humanized (HGF) mouse models; melanoma cell lines. **Phenotype recapitulation:** strong for the melanoma/UV/MAPK axis and both tumor-suppressor pathways. **Limitations:** the combined melanoma + neural-system tumor syndromic phenotype and ARF-specific exon-1β human lesions are incompletely modeled; human-specific p14ARF vs mouse p19ARF differences exist. **Resources:** MGI (mouse *Cdkn2a*), IMPC/KOMP, IMSR.

Mechanistic Model / Interpretation

The unifying model is a **dual-tumor-suppressor, two-hit, two-branch** architecture centered on a single genetic locus:

Feature	p16INK4A branch	p14ARF branch
Encoded by	*CDKN2A* exon 1α	*CDKN2A* exon 1β
Pathway	CDK4/6 → Rb → E2F	MDM2 → p53
Biological brake	Oncogene-induced senescence / G1 arrest	p53-dependent senescence & apoptosis
Principal tumor	Cutaneous melanoma	Astrocytoma / neural-system tumors
Key environmental modifier	UV radiation	— (intrinsic genotoxic stress)
Representative evidence	PMID: 27872202, PMID: 15460900	PMID: 11136714, PMID: 29665672

Large germline 9p21 deletions that disrupt **both** proteins produce the **full melanoma + neural-tumor phenotype**, whereas **ARF-specific (exon 1β) lesions** particularly implicate p14ARF loss in the neural component—the pivotal human genetic insight of this syndrome (PMID: 11136714, PMID: 12019208). Both branches require a somatic **second hit** (homozygous *CDKN2A/B* deletion) and cooperate with MAPK-pathway oncogenes (BRAF/RAS). The syndrome's combined skin-and-nervous-system targeting reflects the **shared neural-crest/neuroectodermal origin** of melanocytes, astrocytes, and Schwann cells.

Evidence Base

PMID	Role in this report	Evidence type
9622062	9p21 lesion disrupting p16/p19 segregates with MAS (MIM 155755)	Human kindred
11136714	ARF-specific exon 1β germline deletion causes neural-tumor component	Human kindred
35904890	CDKN2A exon 1B deletion → melanoma + neural tumors	Human kindred
12019208	ARF haploinsufficiency; three-hit cooperation	Human kindred
15937071	p14ARF exon 1β G16D in CMM+NST family	Human cohort
28699883	Two-hit somatic homozygous CDKN2A/B deletion in MAS astrocytomas	Human case/ tumor
37585199	Prevalence & tumor spectrum; NF1-like phenotype	Biobank cohort
21150883	Penetrance (pancreatic 58%, melanoma 39%); smoking HR 25.8	Human cohort
27872202	p16-mediated senescence barrier to melanoma	In vitro screen
15460900	p16INK4a–Rb molecular mechanism	Review
29665672	ARF–MDM2–p53 axis	Review
21200141	p14ARF/CDKN2A deletion disables p53 in glioma	Review/in vitro
30218143	Carrier melanoma phenotype (younger, multiple, thinner)	Human cohort
32221274	Multiple primary melanomas; CDKN2A/MITF variants	Human cohort
32482799	AD inheritance; p16-Leiden founder	Registry
22636603	p16-Leiden nomenclature; FPC families	Human cohort
35658523	Pancreatic surveillance yields resectable PDAC	Cohort
36785917	Surveillance cost-effectiveness & survival gain	Modeling
16462187	MC1R modifier; UV × genotype penetrance	Review
31371819	Counseling reduces measured UV exposure	Prospective study
23371019	UV dysregulates CDKN2A-het fibroblast transcriptome	In vitro
12406321	Mouse: both Ink4a/Rb & Arf/p53 pathways in melanoma	Mouse

PMID	Role in this report	Evidence type
42849920	Adjuvant therapy improves melanoma survival	Clinical cohort
26183925	CDK4/6 inhibition rational in CDKN2A-null tumors	Preclinical
38287743	Lifetime risk 52–84%; testing yield 10–40%	Review
42630077	Homozygous CDKN2A/B deletion as WHO grade-4 criterion	Review

Limitations and Knowledge Gaps

1. **Rarity and ascertainment bias.** MAS is extremely rare; most data come from small kindreds and registries, giving wide confidence intervals (e.g., melanoma penetrance 95% CI 0–80%). Prevalence estimates span a >4-fold range (1:39,149–1:170,503).
2. **Incomplete ARF-specific genotype–phenotype resolution.** Whether isolated p14ARF loss is *sufficient* for the neural-tumor phenotype, or whether complete locus disruption / additional loci are required, remains partly unresolved (PMID: 15937071).
3. **No syndrome-specific QoL, mortality, or natural-history data.** Outcomes are inferred from component-tumor literature rather than MAS-specific cohorts.
4. **Model gaps.** Mouse models capture the melanoma/MAPK/UV axis well but do not fully recapitulate the combined syndromic neural + cutaneous phenotype or human ARF exon-1β biology.
5. **No animal natural-disease analog** was identified.
6. **Modifier landscape** (beyond MC1R/MITF) and epigenetic contributions to the neural branch are underexplored.

Proposed Follow-up Experiments / Actions

1. **ARF-isoform-specific mouse models** (conditional p19ARF-only deletion in neural-crest lineages) to test sufficiency of ARF loss for astrocytoma/schwannoma.
2. **Multi-omic profiling of MAS tumors** (WGS + methylation + single-cell) to map the somatic second-hit landscape and MAPK cooperation across the melanoma and neural branches.

3. **Prospective MAS-specific natural-history registry** capturing penetrance by tumor type, age, sex, and modifier genotype (MC1R, MITF), with QoL instruments.
4. **Clinical trials of CDK4/6 inhibitors** (± MAPK or IDH inhibitors) in CDKN2A-null MAS-associated tumors to validate the precision-oncology rationale (PMID: 26183925).
5. **Optimize combined surveillance protocols** (skin + brain MRI + pancreatic MRI/EUS) and evaluate cost-effectiveness of adding neuroimaging for carriers with ARF-affecting variants.
6. **Behavioral-intervention trials** to sustain UV-reduction gains after genetic counseling (PMID: 31371819).

Report compiled from 13 confirmed findings across 5 investigation iterations and 64 reviewed papers. Evidence types span human kindreds/cohorts, biobank data, in vitro studies, mouse models, and reviews.
